Ranked SIMEX Corrected Student Growth Percentiles: Why they are needed and preliminary implementation with Georgia Milestones data.

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1 The Measurement Error Problem and Corrections

Measurement error (ME) is an inherent component of all standardized tests, and the impact that ME can have when test score data are used to compute student growth and teacher/school evaluation measures has been the focus of a growing body of academic research. Specifically in the area of Student Growth Percentile (SGP) measures of student progress, ME has been found to create bias that can disadvantage students with lower prior achievement and vice versa. This bias is transferred to aggregate measures of educator effectiveness when a disproportionate number of students with relatively low/high prior achievement are concentrated in a classroom or school (Lockwood & Castellano, 2015; Shang, VanIwaarden, & Betebenner, 2015).

Researchers have proposed several useful methods for correcting the effects of ME. The use of Simulation/Extrapolation (SIMEX) techniques has been found to effectively eliminate the ME induced bias related to prior achievement in SGPs (Shang et al., 2015), and this method is a currently available for SGP calculation in the `SGP` package (Betebenner, VanIwaarden, Domingue, & Shang, 2017) for the `R` statistical program. Currently several states utilize the SIMEX corrected measures in their student growth modeling and evaluation policies.

However, the SIMEX corrected SGPs also have technical limitations. At an individual level, the corrected SGPs have larger errors than the uncorrected, or "standard", SGPs (McCaffrey, Castellano, & Lockwood, 2015; Shang et al., 2015). McCaffrey, et al. (2015) first suggested that ranking the SIMEX SGP values may present a possible alternative that would have the beneficial properties of both SGP estimate types. Castellano and McCaffrey (2017) recently investigated the properties of the percentile ranked SIMEX SGP (RS-SGP) at the aggregate level for MSGP estimates of educator effectiveness. They found that the majority of the error variance in standard MSGP values is due to "sampling variability" (i.e. a classroom is considered a sample of all possible students), but that a substantial amount was also due to bias caused by ME. SIMEX correction can remove much of this bias initially. By subsequently taking the percentile ranks of the SIMEX corrected values and then aggregating these percentile ranks, the excess variance from the SIMEX estimation is removed. Furthermore they found that, at the individual level, the distribution of the RS-SGPs is also more uniformly distributed (similar to the standard SGP) rather than the SIMEX SGPs typical U-shaped distribution. The uniform distribution of the individual SGP values is a desirable characteristic because it suggests that the full range of SGP growth values (1-99) is equally likely to be attained.

Given the potential promise of RS-SGP, it is now calculated along with the SIMEX values in the `SGP` package[1]. Further insights from that implementation are discussed next.

2 SGP Package Implementation of Ranked SIMEX SGP

In their study, Castellano and McCaffrey report simply taking the percentile rank of the computed SIMEX SGP values to get the RS-SGP. However, unlike the SIMEX SGP values computed through data simulations in the `SGP` package, they compute their values using a closed-form equation. This produces continuous SIMEX SGP values, which allow for a more detailed ranking than using the integer values computed in the `SGP` package. Although their process helps to better understand the theoretical groundings of the various SGP estimates, it

[1]SGP versions 1.7-0.0 and later

is only appropriate under particular assumptions about the data and ME structures that do not hold in the real-world situations.

Without a continuous value, the percentile ranking[2] of a set of numbers that is already on a percentile scale does not produce results that differ substantially from the original in absolute value or distribution. Therefore a solution was required in the simulation process that would allow for a more continuous SIMEX SGP to be established. In following up with the authors they suggested that more granular SIMEX SGPs be established in the simulations, however this would require the already computationally and time intensive process to take 10 times longer. Furthermore, previously calculated SIMEX values would no longer be reproducible. A simpler solution was used that allows the estimated SIMEX values to be placed on a $1/8^{th}$ interval by calculating arithmetic midpoints between each percentile's predicted score values[3]. This resulted in RS-SGP values that were more uniformly distributed in initial tests with real and simulated data.

Figure 1 shows the distribution of the three types of SGP estimates for 2016 8^{th} Grade Mathematics SGP analyses in Georgia: uncorrected ("standard"), SIMEX corrected and Ranked SIMEX. Note that these results are from analyses that use up to three years of data (two prior and the current year), which the authors indicate will also greatly reduce ME bias.

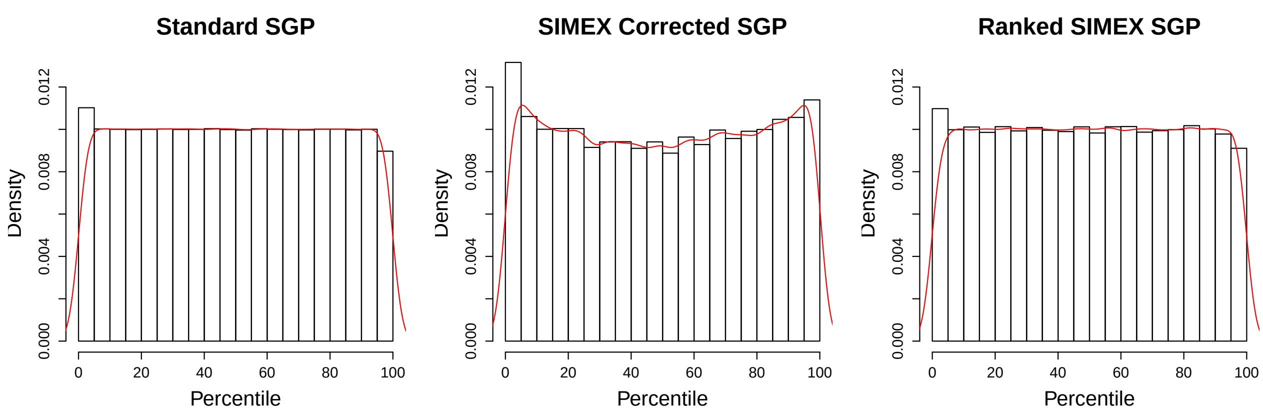

Figure 1: Comparison of the Uniformity of Distributions for 8^{th} Grade Mathematics Estimates.

By definition, the standard SGP is uniformly distributed *given any prior test score*, suggesting that any level of growth is equally likely regardless of prior achievement. This is a critical distinction, and Castellano and McCaffrey do not discuss the conditional uniformity of the RS-SGP. We find that this uniformity is not met in either the application of the closed-form equations to simulated data or in our initial tests with real data in the `SGP` package, although the RS-SGP distribution is much closer to uniform than that of the SIMEX SGPs.

The following figures are "Goodness of Fit" charts that are produced using the `SGP` package for each of the three SGP estimate types, and they can help investigate the SGP distribution in more detail. The "Student Growth Percentile Range" panel at bottom left shows the empirical distribution of SGPs given prior scale score deciles in the form of a 10 by 10 cell grid.

[2]Calculated as (rank(SIMEX SGP)/N) x 100 where N is the number of students. The result is rounded to the nearest integer.

[3]SGP estimates are found by predicting 100 scores for each student - one for each percentile. The position (1-99) of the predicted score that is closest to a student's observed score is their estimated SGP.

Percentages of student growth percentiles between the 10^{th}, 20^{th}, 30^{th}, 40^{th}, 50^{th}, 60^{th}, 70^{th}, 80^{th}, and 90^{th} percentiles were calculated based upon the empirical decile of the cohort's prior year scaled score distribution. Perfect uniform distribution conditional on prior score would be indicated by a "10" in each cell. Deviations from perfect fit are indicated by red and blue shading. The further above 10 the darker the red, and the further below 10 the darker the blue. The bottom right panel of each plot is a Q-Q plot which compares the observed distribution of SGPs with the theoretical (uniform) distribution. An ideal plot here will show black step function lines that do not deviate from the ideal, red line which traces the 45 degree angle of perfect fit (as is seen here in the first plot for the standard SGP).

These plots display typical distributions of each SGP variant from the same 2016 8^{th} Grade Mathematics SGP analyses as depicted above. The Standard SGPs are nearly perfectly distributed conditional upon prior achievement. The SIMEX and, to a lesser extent, RS-SGP distributions are skewed towards higher percentiles at the lower levels of achievement and lower growth for the higher prior achievement deciles.

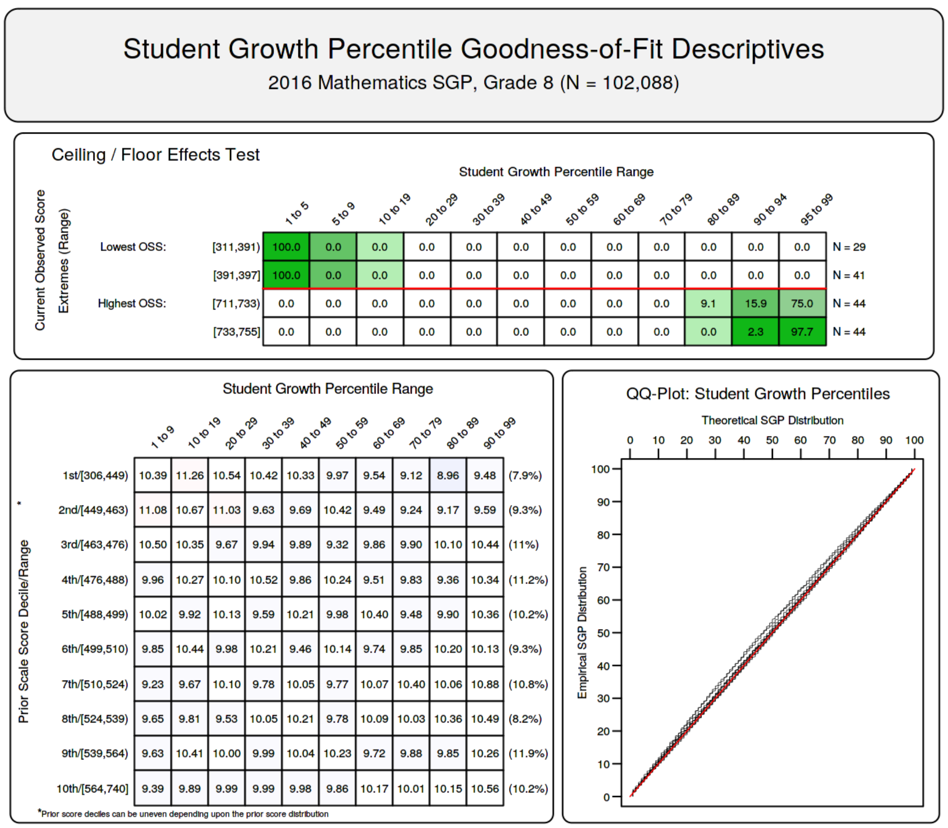

Figure 2: Goodness of Fit Plot for 2016 ***Standard*** 8^{th} Grade Mathematics SGPs.

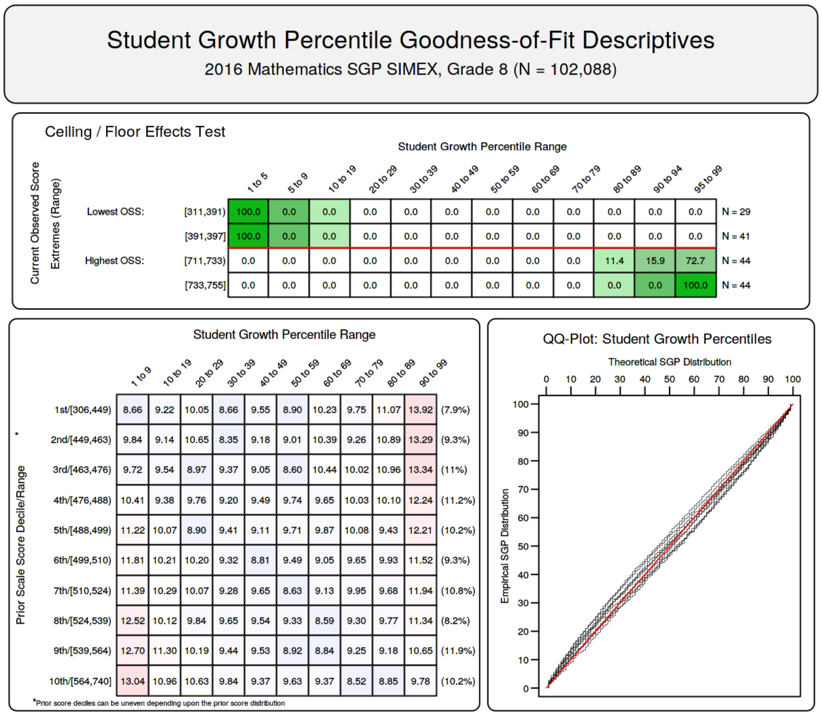

Figure 3: Goodness of Fit Plot for 2016 ***SIMEX*** 8^{th} Grade Mathematics SGPs.

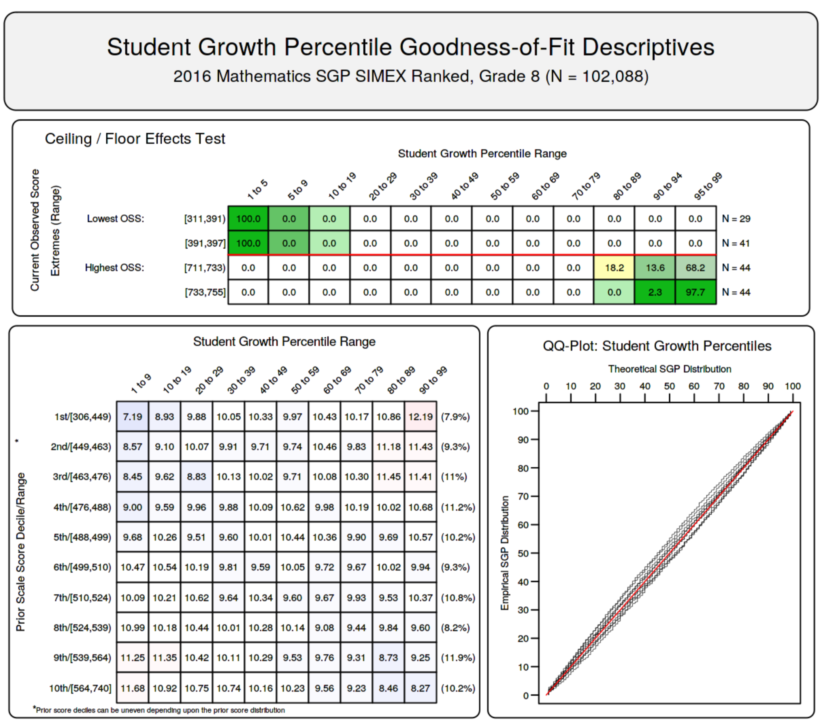

Figure 4: Goodness of Fit Plot for 2016 ***Ranked SIMEX*** 8^{th} Grade Mathematics SGPs.

3 Relationship of Ranked SIMEX with Prior Student Achievement

An important consequence of the typical SIMEX and RS-SGP conditional distributions is that a negative correlation is created between them and prior test scores. This is true at the student level and also translates to school and teacher aggregations. These negative relationships are indicative of the reduction in ME induced bias. The following tables show the results for the End-of-Grade (EOG) and End of Course (EOC) analyses for Georgia from 2016. As can be seen, the SIMEX and RS-SGP correlations are nearly identical in both settings. This is unsurprising as the maximum differences between the two values are between -3 and 3 with near-zero averages for all grade-by-subject specific analyses (see subsequent sections for more details on the observed differences between SIMEX and RS-SGP).

3.1 EOG Subjects

Table 1: Student Level Correlations between Prior Standardized Scale Score and 1) Current Scale Score, 2) Standard SGP, 3) SIMEX SGP and 4) Ranked SIMEX SGP.

Content Area	Grade	$r_{TestScores}$	$r_{Standard}$	r_{SIMEX}	r_{Ranked}	N Size
ELA	4	0.83	0.00	-0.12	-0.12	123,767
	5	0.83	0.01	-0.07	-0.07	122,053
	6	0.84	0.01	-0.07	-0.07	119,853
	7	0.84	0.02	-0.07	-0.07	118,962
	8	0.83	0.02	-0.07	-0.07	120,474
Mathematics	4	0.84	0.00	-0.10	-0.10	124,313
	5	0.84	0.01	-0.07	-0.07	122,680
	6	0.84	0.01	-0.06	-0.06	120,376
	7	0.87	0.02	-0.07	-0.07	118,986
	8	0.81	0.02	-0.06	-0.06	102,088
Science	4	0.79	0.00	-0.10	-0.10	124,092
	5	0.80	0.00	-0.07	-0.07	122,488
	6	0.80	0.00	-0.08	-0.08	120,164
	7	0.83	0.00	-0.07	-0.07	119,107
	8	0.78	0.00	-0.06	-0.06	94,694
Social Studies	4	0.77	0.00	-0.09	-0.09	123,446
	5	0.80	0.00	-0.08	-0.08	121,783
	6	0.80	0.00	-0.08	-0.08	119,550
	7	0.84	0.00	-0.08	-0.08	118,545
	8	0.82	0.00	-0.06	-0.06	119,813

3.2 EOC Subjects

Each EOC test subject is analyzed using more than one sequence of prior subjects, grades and years, and these unique progressions are disaggregated in Tables 2 and 3 using the most recent prior available for each norm group (although more prior years' scores are used in SGP calculations when available). These correlations between current and prior scale score are notably lower than in the grade level norm groups, and overall lower correlations may be expected in EOC subjects due to the change in specific subject from one course to the next.

As expected, the relationships between growth and prior achievement at the student level reported in Tables 2 and 3 are still non-existent for standard SGPs and slightly negative after SIMEX correction and SIMEX ranking. SIMEX and RS-SGP correlations are indistinguishable for EOCs as well.

Table 2: EOC Student Level Correlations between Prior Standardized Score and 1) Current Scale Score, 2) Standard SGP, 3) SIMEX SGP and 4) Ranked SIMEX SGP - Disaggregated by Norm Group.

Content Area	Most Recent Prior	$r_{TestScores}$	$r_{Standard}$	r_{SIMEX}	r_{Ranked}	N Size
Grade 9 Lit	2015 ELA Grade 7	0.75	0.02	-0.10	-0.10	4,075
	2015 ELA Grade 8	0.82	0.02	-0.07	-0.07	115,378
	2015 Grade 9 Lit	0.63	0.00	-0.09	-0.09	5,279
American Lit	2013 Grade 9 Lit	0.78	0.00	-0.10	-0.10	2,500
	2014 Grade 9 Lit	0.79	0.02	-0.07	-0.07	95,761
	2015 American Lit	0.65	0.00	-0.09	-0.09	4,596
	2015 Grade 9 Lit	0.77	-0.01	-0.10	-0.10	8,176
US History	2012 Soc Studies Grade 8	0.66	-0.01	-0.07	-0.07	1,909
	2013 Soc Studies Grade 8	0.72	0.00	-0.08	-0.08	84,784
	2014 Soc Studies Grade 8	0.74	0.00	-0.09	-0.09	10,283
	2015 Economics	0.76	-0.01	-0.10	-0.10	4,389
	2015 US History	0.67	0.00	-0.10	-0.10	2,590
	2016 Economics	0.72	0.00	-0.08	-0.08	2,081
Economics	2012 Soc Studies Grade 8	0.73	0.00	-0.08	-0.08	3,309
	2013 Soc Studies Grade 8	0.71	0.00	-0.08	-0.08	2,858
	2014 Soc Studies Grade 8	0.73	0.00	-0.11	-0.11	3,146
	2014 US History	0.70	0.00	-0.06	-0.06	5,832
	2015 Soc Studies Grade 8	0.74	0.00	-0.11	-0.11	2,225
	2015 US History	0.71	0.00	-0.08	-0.08	88,780
	2016 Economics	0.66	-0.01	-0.10	-0.10	1,518

Table 3: EOC Student Level Correlations between Prior Standardized Score and 1) Current Scale Score, 2) Standard SGP, 3) SIMEX SGP and 4) Ranked SIMEX SGP - by Norm Group (*Continued*).

Content Area	Most Recent Prior	$r_{TestScores}$	$r_{Standard}$	r_{SIMEX}	r_{Ranked}	N Size
Coordinate Algebra	2015 Coord Algebra	0.71	0.00	-0.11	-0.11	4,189
	2015 Math Grd 7	0.73	0.01	-0.10	-0.10	3,148
	2015 Math Grd 8	0.78	0.01	-0.07	-0.07	19,544
Analytic Geometry	2014 Coord Algebra	0.57	0.00	-0.08	-0.08	1,831
	2015 Analytic Geom	0.58	0.00	-0.10	-0.10	6,792
	2015 Coord Algebra	0.75	0.02	-0.07	-0.07	84,634
Algebra I	2015 Math Grd 7	0.76	0.00	-0.10	-0.10	15,646
	2015 Math Grd 8	0.80	0.02	-0.06	-0.06	75,621
Geometry	2015 Coord Algebra	0.75	0.02	-0.07	-0.07	33,006
Physical Science	2014 Biology	0.71	0.02	-0.05	-0.05	4,951
	2014 Science Grd 8	0.69	-0.01	-0.07	-0.07	3,236
	2015 Biology	0.73	-0.01	-0.08	-0.08	36,140
	2015 Physical Science	0.58	0.00	-0.11	-0.11	2,944
	2015 Science Grd 7	0.73	0.00	-0.07	-0.07	26,138
	2015 Science Grd 8	0.78	0.00	-0.08	-0.08	18,411
Biology	2014 Physical Science	0.84	0.00	-0.09	-0.09	2,753
	2014 Science Grd 8	0.74	-0.01	-0.07	-0.07	12,554
	2015 Biology	0.67	0.00	-0.10	-0.10	5,015
	2015 Physical Science	0.69	0.01	-0.09	-0.09	39,489
	2015 Science Grd 8	0.77	-0.03	-0.10	-0.10	60,608

3.3 Schools

We must also consider the impact the SIMEX ranking has on aggregated SGPs since they are used for school and teacher accountability in Georgia. The following tables looks at the correlations between school level aggregations (averages) of prior student achievement and growth from analyses for Georgia. Three years of results are provided.

Table 4: 2014 to 2016 School Level EOG Correlations between Mean Prior Standardized Scale Score and Aggregate SGPs by Content Area.

Content Area	Year	R Mean SGP	R Mean SIMEX	R Mean Ranked SIMEX	N
ELA	2014	0.55	0.35	0.35	1,734
	2015	0.42	0.26	0.26	1,743
	2016	0.58	0.43	0.43	1,746
Mathematics	2014	0.45	0.32	0.32	1,734
	2015	0.46	0.36	0.36	1,743
	2016	0.51	0.41	0.41	1,746
Science	2014	0.41	0.25	0.25	1,737
	2015	0.40	0.28	0.28	1,746
	2016	0.50	0.39	0.39	1,745
Social Studies	2014	0.35	0.20	0.20	1,737
	2015	0.34	0.20	0.20	1,746
	2016	0.44	0.33	0.33	1,746

Table 5: 2014 to 2016 School Level EOC Correlations between Mean Prior Standardized Scale Score and Aggregate SGPs by Content Area.

Content Area	Year	R Mean SGP	R Mean SIMEX	R Mean Ranked SIMEX	N
Grade 9 Lit	2014	0.44	0.21	0.21	468
	2015	0.33	0.21	0.21	485
	2016	0.55	0.43	0.43	493
American Lit	2014	0.49	0.26	0.26	425
	2015	0.31	0.18	0.18	425
	2016	0.60	0.51	0.51	433
US History	2014	0.11	0.01	0.01	421
	2015	0.14	0.05	0.05	422
	2016	0.27	0.18	0.18	428
Economics	2014	0.15	0.06	0.06	416
	2015	0.12	0.04	0.04	424
	2016	0.38	0.29	0.29	425
Coordinate Algebra	2014	0.40	0.24	0.24	639
	2015	0.36	0.22	0.22	711
	2016	0.33	0.22	0.22	206
Analytic Geometry	2014	0.48	0.32	0.32	421
	2015	0.45	0.33	0.33	431
	2016	0.53	0.42	0.42	332
Algebra I	2016	0.35	0.24	0.24	560
Geometry	2016	0.59	0.48	0.48	122
Physical Science	2014	0.30	0.19	0.19	457
	2015	0.33	0.24	0.24	529
	2016	0.34	0.27	0.27	574
Biology	2014	0.21	0.08	0.08	426
	2015	0.30	0.20	0.20	436
	2016	0.33	0.23	0.23	438

4 Differences between SIMEX and Ranked SIMEX SGPs

Given that the correlations between prior achievement and growth are essentially identical for SIMEX and RS-SGPs, and yet there are visible differences in their distributions and model fit charts, we now investigate the absolute differences between the two values at the student and school levels.

4.1 SGP Change at Individual Student Level

When replacing SIMEX with ranked SIMEX, students with higher SIMEX SGP tend to receive lower ranked SIMEX SGP, students with lower SIMEX SGP tend to receive higher ranked SIMEX SGP. The correlation between SGP difference and SIMEX SGP is -0.82. Most of the students have SGP change within 2 percentiles.

Table 6: The Change of SGP (Ranked SIMEX - SIMEX) by EOG Subject and MeanGP SIMEX.

Subject	SGP Difference	Mean SIMEX SGP	N
ELA	-3	84.56	4,903
	-2	82.53	115,599
	-1	74.94	129,685
	0	50.71	93,362
	1	25.93	124,764
	2	18.52	125,827
	3	15.09	10,969
Mathematics	-4	85.00	19
	-3	82.50	8,834
	-2	81.21	116,896
	-1	73.41	132,958
	0	46.01	103,661
	1	22.51	152,706
	2	16.93	72,888
	3	19.45	481
Science	-3	82.27	974
	-2	81.08	87,952
	-1	75.75	153,206
	0	47.13	117,345
	1	22.71	160,754
	2	17.95	59,845
	3	17.57	469
Social Studies	-3	78.44	590
	-2	80.80	95,647
	-1	75.55	160,955
	0	46.18	125,384
	1	21.93	167,644
	2	16.95	52,626
	3	17.57	291

Table 7: The Change of SGP (Ranked SIMEX - SIMEX) by EOC Subject and MeanGP SIMEX.

Subject	SGP Difference	Mean SIMEX SGP	N
Grade 9 Lit	-3	84.19	1,229
	-2	83.39	24,230
	-1	75.51	23,621
	0	51.63	21,654
	1	25.16	26,820
	2	18.56	26,322
	3	16.40	856
American Lit	-5	95.00	6
	-4	90.36	184
	-3	84.93	3,206
	-2	82.28	26,494
	-1	70.17	19,128
	0	47.46	17,546
	1	23.81	27,111
	2	16.51	16,456
	3	14.97	891
	4	15.00	11
US History	-3	79.48	42
	-2	81.69	8,304
	-1	76.15	34,751
	0	43.85	29,461
	1	21.07	30,119
	2	19.75	3,351
	3	34.00	8
Economics	-4	70.44	16
	-3	76.48	182
	-2	79.48	4,256
	-1	75.06	39,196
	0	43.38	34,855
	1	20.38	28,252
	2	19.81	911

Table 8: The Change of SGP (Ranked SIMEX - SIMEX) by EOC Subject and MeanGP SIMEX (Continued).

Subject	SGP Difference	Mean SIMEX SGP	N
Coordinate Algebra	-3	88.11	132
	-2	82.42	3,935
	-1	77.28	6,543
	0	47.85	5,603
	1	22.82	7,710
	2	17.66	2,913
	3	20.76	45
Analytic Geometry	-3	82.38	945
	-2	82.56	13,398
	-1	76.40	22,639
	0	46.66	21,341
	1	22.32	26,164
	2	17.29	8,645
	3	16.11	125
Algebra I	-4	79.00	4
	-3	84.74	310
	-2	83.60	13,400
	-1	75.83	23,323
	0	46.04	19,656
	1	22.45	25,781
	2	18.07	8,790
	3	21.33	3
Geometry	-4	87.79	24
	-3	84.23	1,120
	-2	80.91	5,696
	-1	74.26	7,190
	0	50.15	5,323
	1	22.73	8,659
	2	17.73	4,436
	3	13.80	558

Table 9: The Change of SGP (Ranked SIMEX - SIMEX) by EOC Subject and MeanGP SIMEX (Continued).

Subject	SGP Difference	Mean SIMEX SGP	N
Physical Science	-3	82.81	16
	-2	82.45	5,699
	-1	76.95	29,913
	0	45.79	24,084
	1	21.80	30,114
	2	19.10	1,991
	3	19.00	3
Biology	-3	85.43	606
	-2	80.74	22,085
	-1	75.25	26,934
	0	48.13	21,032
	1	23.12	29,226
	2	18.53	20,181
	3	16.25	355

4.2 Mean SGP Change at School Level

Most of the schools have mean student growth percentile (MSGP) change within 2 percentiles when switching from SIMEX to Ranked SIMEX.

Table 10: Descriptive statistics of school MGP change when switching from SIMEX to ranked SIMEX SGP.

Subject	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	SD	N
ELA	-1.56	-0.22	-0.06	-0.07	0.09	0.67	0.24	1,746
Mathematics	-1.17	-0.07	0.16	0.14	0.36	1.29	0.31	1,746
Science	-0.81	-0.11	0.06	0.05	0.21	1.06	0.24	1,745
Social Studies	-0.97	-0.06	0.11	0.11	0.29	1.03	0.27	1,746
Grade 9 Lit	-1.06	-0.24	-0.05	-0.04	0.14	1.16	0.32	493
American Lit	-1.31	-0.13	0.13	0.10	0.34	1.06	0.35	433
US History	-0.93	-0.12	0.09	0.08	0.27	0.72	0.29	428
Economics	-0.60	-0.01	0.14	0.13	0.31	0.80	0.23	425
Coordinate Algebra	-0.82	-0.26	0.03	0.08	0.39	1.39	0.48	206
Analytic Geometry	-0.71	-0.15	0.04	0.05	0.25	0.97	0.30	332
Algebra I	-0.90	-0.19	0.13	0.14	0.44	1.41	0.45	560
Geometry	-0.89	-0.26	-0.07	0.00	0.27	1.04	0.38	122
Physical Science	-0.67	-0.15	0.06	0.08	0.28	1.10	0.33	574
Biology	-1.06	-0.20	-0.01	0.00	0.24	0.93	0.35	438

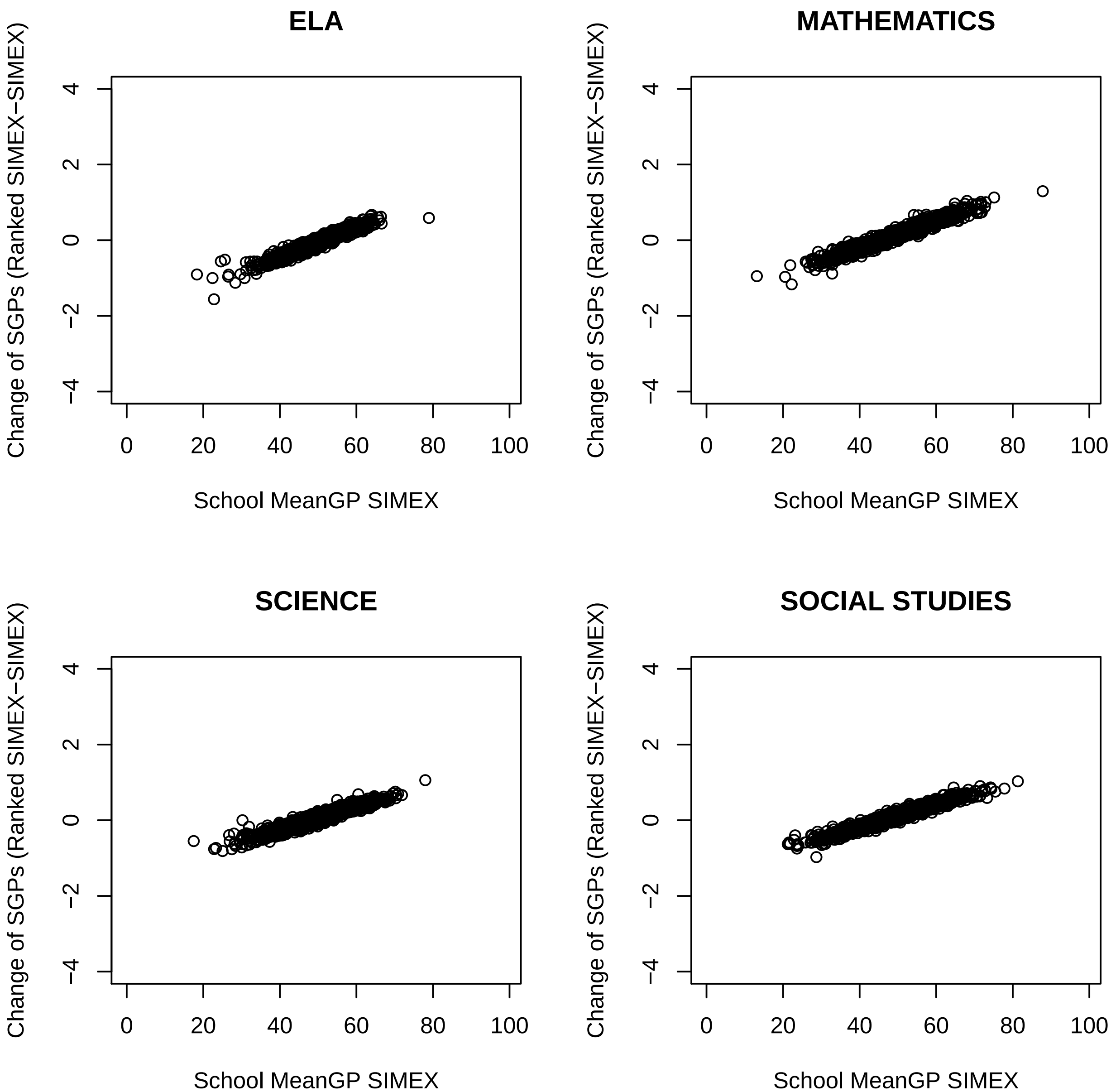

Figure 5: School MGP change (MGP Ranked SIMEX- MGP SIMEX) by EOG Subjects.

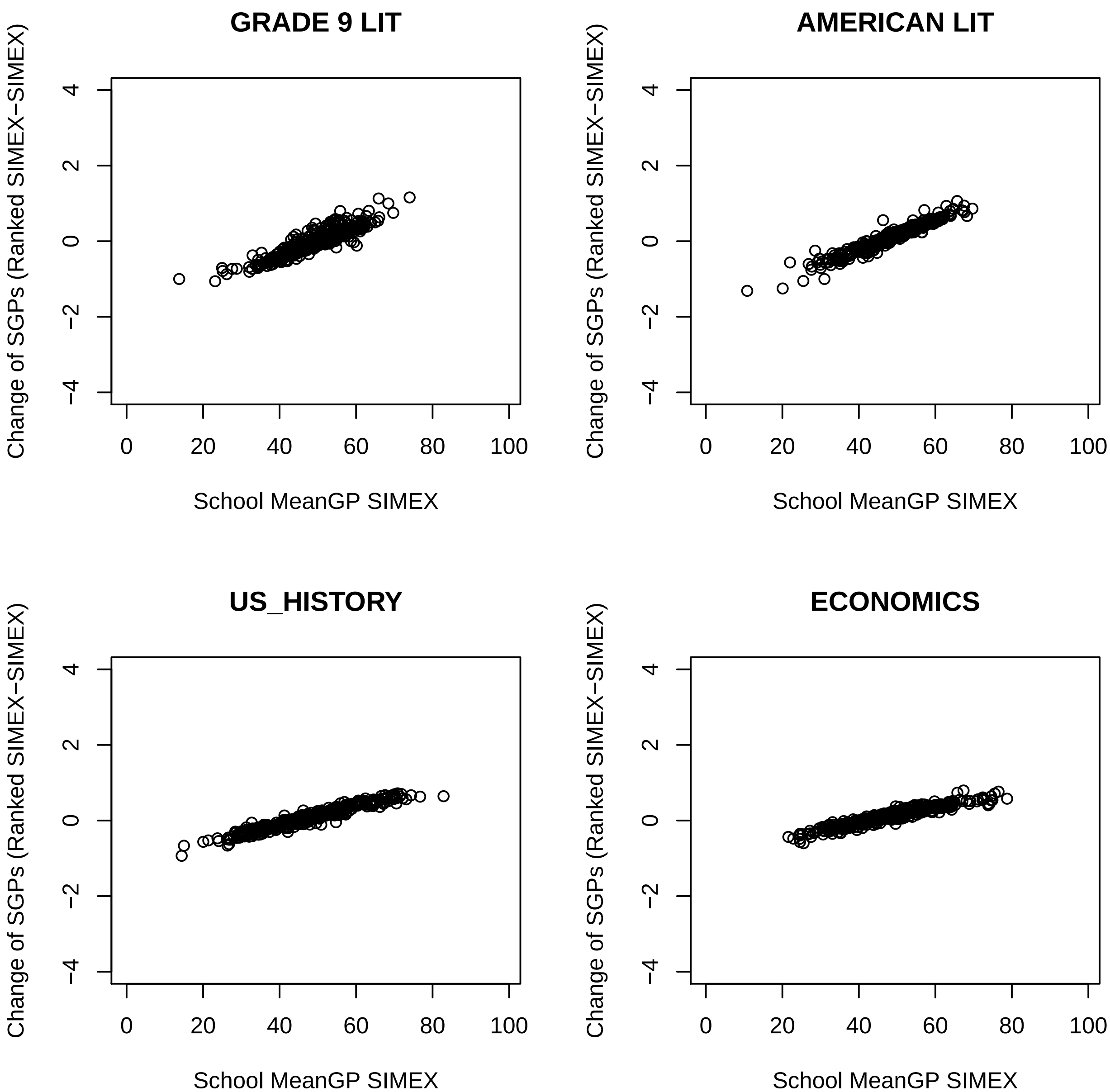

Figure 6: School MGP change (MGP Ranked SIMEX- MGP SIMEX) by EOC Subjects.

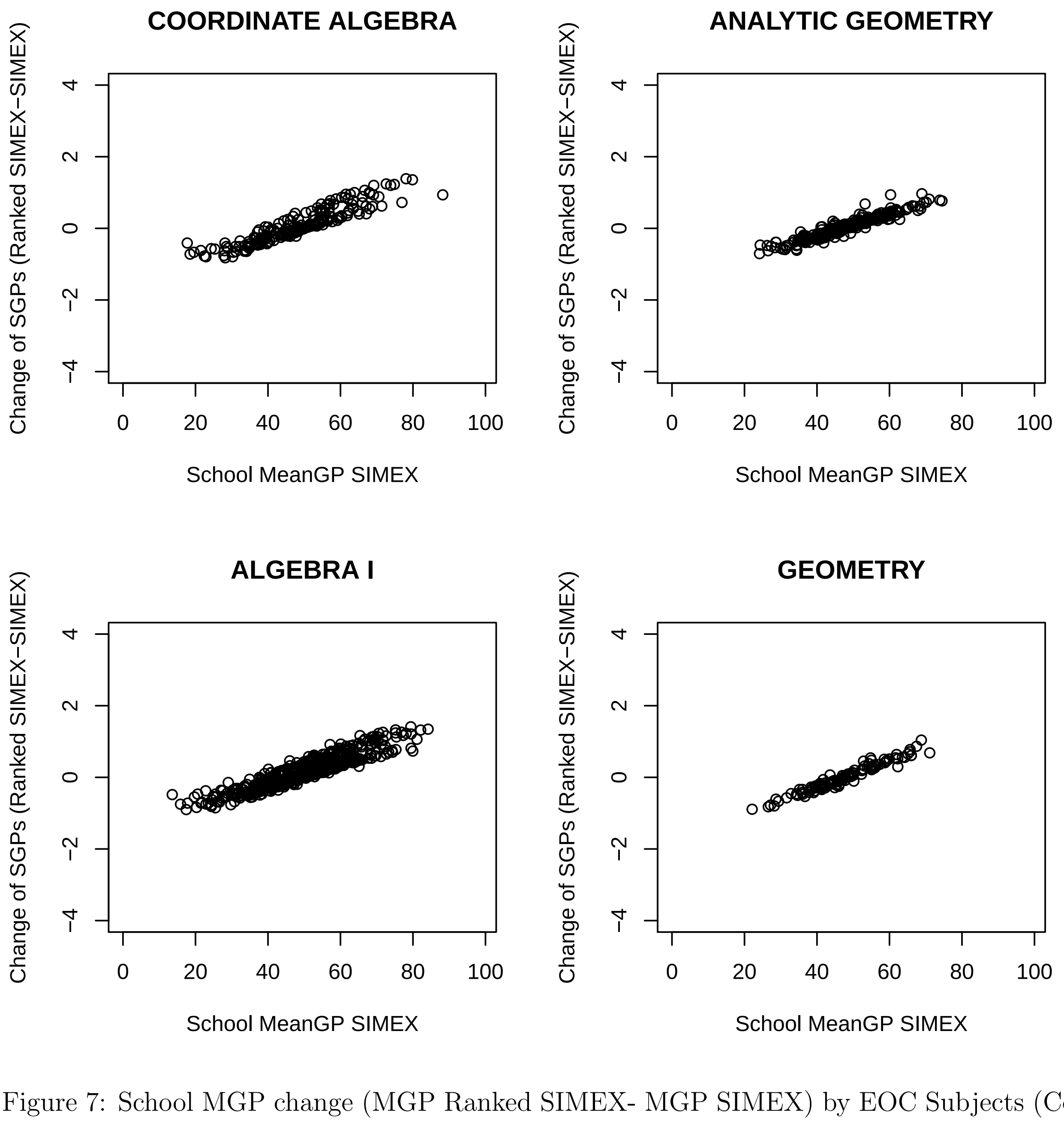

Figure 7: School MGP change (MGP Ranked SIMEX- MGP SIMEX) by EOC Subjects (Continued).

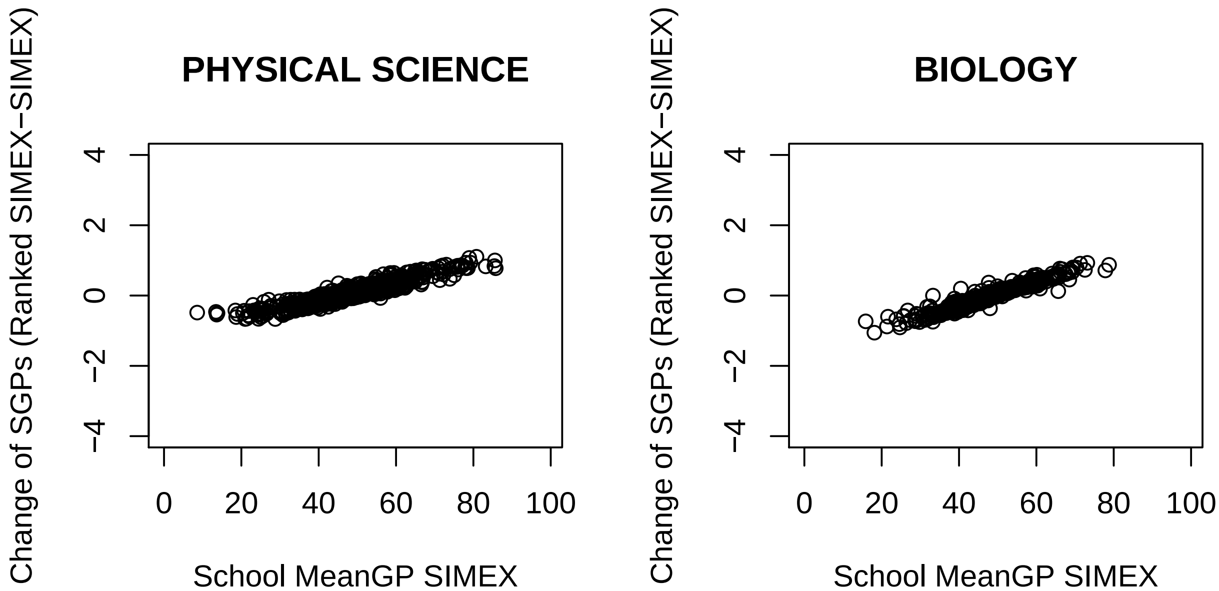

Figure 8: School MGP change (MGP Ranked SIMEX- MGP SIMEX) by EOC Subjects (Continued).

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